
ARTICLES

Determination of the Average Molecular Weight of Hydroxyethyl Starches by Diffusion-Ordered NMR Spectroscopy

N. E. Kuz'mina, S. V. Moiseev*, V. I. Krylov, V. A. Yashkir, and V. A. Merkulov

*Scientific Center for Expertise of Medical Products, Ministry of Health of the Russian Federation,
Petrovskii bul'var 8, Moscow, 127051 Russia*

**e-mail: MoiseevSV@expmed.ru*

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Abstract—A correlation equation is obtained for the evaluation of the average molecular weight (MW) of hydroxyethyl starches from the self-diffusion coefficient D measured using diffusion-ordered NMR spectroscopy. Correlation equations $MW = cD^{-\alpha}$ are compared for hydroxyethyl starches, dextrans, and pullulans. It has been demonstrated that different MWs correspond to constant D values of uncharged polysaccharides with the similar type of nonvalent interactions and different degrees of branching.

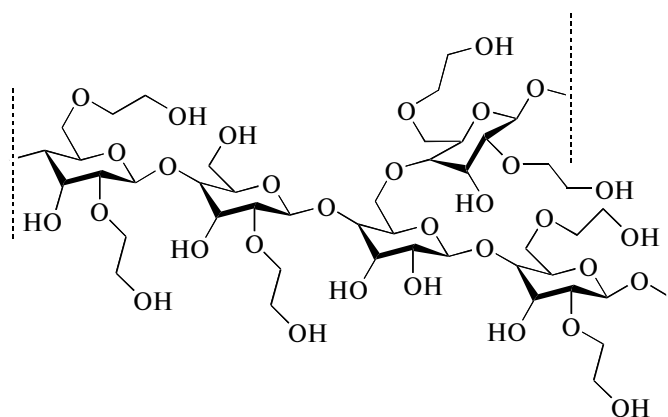
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INTRODUCTION

Hydroxyethyl starches (HES) are efficient colloid plasma substituents, which are widely used in clinical practice for the prevention and treatment of hypovolemia in surgery, traumatology, combustiology, and intensive therapy over the last 50 years [1–3]. The molecular structure of HES represents a polydisperse

branched polymer amylopectin consisting of α -D-glucopyranose residues modified with hydroxyethyl groups (Scheme). The linear section of the HES macromolecules contains $\alpha(1 \rightarrow 4)$ bonds, branching are formed mostly by $\alpha(1 \rightarrow 6)$ bonds [4, 5]. The side branches of the macromolecule contain from 15 to 45 glucose residues.



Fragment of the molecular structure of HES.

The important characteristics of HES affecting their pharmacokinetic properties are the average molecular weight; molar substitution (MS), which is the ratio of the number of hydroxyethyl groups to the total number of glucose molecules; and the ratio C2/C6, which characterizes the ratio of the number of hydroxyethyl groups at the second carbon atom to the

number of hydroxyethyl groups at the sixth carbon atom in the glucopyranose cycle [6–8]. MW determines the time of the HES presence in the blood flow. HES macromolecules and fragments of their enzymatic hydrolysis penetrating through the kidney barrier (45–60 kDa) are rapidly eliminated from the body [7]. Hydroxyethylation slows down the enzymatic

hydrolysis of the HES molecules, thus extending the time of circulation of the preparation in the blood flow [8]. A similar effect is made by the high C2/C6 ratio: the higher the C2/C6 ratio, the slower the decomposition of HES [6–8].

The determination of the quantitative characteristics of HES (MW, MS, and C2/C6) is a rather complex and laborious problem; in the pharmacopeia analysis it is solved using a complex of chromatographic methods: gel-permeation chromatography (GPC) (MW determination), and GC and HPLC (determination of MS and C2/C6) [9, 10]. The application of chromatographic methods to the determination of MS and C2/C6 is based on the destruction of the analyzed HES. Therefore, the development of a nondestructive rapid method that ensures the quick and reliable estimation of all quantitative HES properties using one sample with the minimal sample preparation is a problem of current importance. For the development of such a method mostly NMR spectroscopy is used. The quantitative determination of MS and sites of HES substitution was performed based on the data of ^1H and ^{13}C NMR spectra [4, 11–16]. Methods of traditional NMR spectroscopy are inapplicable to the determination of the MWs of polymers. Diffusion-ordered NMR spectroscopy (DOSY), which ensures the measurement of the diffusion of various molecular objects (molecules, macromolecules, molecular complexes, and supramolecular systems) under the impact of a magnetic field gradient [17–19], seems to be the most promising method for this purpose. The DOSY method possesses high potential in the studies of the molecular weights of polymers [20–23], including polysaccharides [24–28]. In practice [26–28], the MWs of polysaccharides within the frameworks of the DOSY method were evaluated using the correlation equation proposed by the authors of [24] for the dependence of the MW of polysaccharides on the self-diffusion coefficient D , which is numerically equal to the mean square shift of a molecule during the time of diffusion t_d [29]. This equation was obtained on the basis of the analysis of linear and low-branched polysaccharides (dextrans and pullulans). It is known that the topology of a polymer substantially affects its diffusion, because branched molecules always have smaller hydrodynamic radii than their linear analogues [30]; therefore the use of the correlation equation for the quantitative evaluation of the MW of HES presented in [24] can lead to incorrect results.

The aim of the present work was to study the effect of branching on the D value of polysaccharides and the evaluation of a correlation relationship between the MW of HES and D taking into account the peculiarities of their steric structures.

EXPERIMENTAL

We used HES standard samples produced by the American Polymer Standard Corporation (United States) with the average molecular weights M_p^1 5.5, 8.9, 16.1, 52.8, 80.0, 121.1, 153.6, 176.1, 201.0, 219.3, 330.2, 374.0, 396.9, 468.7, and 1004.0 kDa (HETA 6K, HETA 9K, HETA 18K, HETA 60K, HETA 88K, HETA 130K, HETA 160K, HETA 175K, HETA 230K, HETA 250K, HETA 337K, HETA 380K, HETA 460K, HETA 600K, and HETA 1300K, respectively). The ^1H NMR and ^1H -DOSY spectra were recorded on an Agilent DD2 NMR System 600 NMR-spectrometer with a 5 mm inverse multinuclear sensor equipped with a gradient coil at 300 K. The analyzed samples (5 mg) were dissolved in 0.5 mL of D_2O (Cambridge Isotope Laboratories Inc.). The solutions obtained were diluted to the concentration 1 mg/mL and transferred into Shigemi NMR tubes with a glass plunger for the minimization of the effects of convection and the heterogeneity of pulses of the magnetic field gradient. For measurements of the self-diffusion coefficient, a DBPPSTE (DOSY Bipolar Pulse Pair Stimulated Echo) sequence was used. Diffusion attenuation curves were obtained at a sequential 15-step linear increase of the amplitude of the magnetic field gradient pulse in the range from 1.82 to 53 G/cm at fixed values of diffusion time Δ (285 ms), δ_{pulse} gradient pulse duration (2.0 ms), and relaxation time $d1$ (5 s).

The results were mathematically processed using the DISCRETE method (Discrete Sum of Exponential Decays). Correlation equation was derived using the MS Excel 2007 software.

RESULTS AND DISCUSSION

HES is a polydisperse polymer; its MW is an average value and depends on the mode of averaging. In polymer chemistry, number average (M_n) and weight average (M_w) molecular weights are distinguished. M_n is obtained by averaging by the number of macromolecules in a polymer; M_w is obtained by averaging by the weight of macromolecules in a polymer. The M_w value is sensitive to high-molecular fractions, while M_n is sensitive to low-molecular weight fractions of polydisperse polymers; therefore, for the characteristic of the average MW of HES we used an averaged parameter representing molecular weight in peak (M_p), determined by chromatographic methods. For narrow-dispersed samples $M_p = (M_w \times M_n)^{0.5}$ [31]. The correlation relationship between the MW and D values is described by a power function [24, 27, 32, 33]:

$$D = cMW^\alpha,$$

¹ M_p is the molecular weight of a peak in its maximum in the determination of the average molecular weight of a polymer by exclusion gel-permeation chromatography.

Table 1. Self-diffusion coefficients of components of aqueous HES solutions found using the DOSY method ($c = 1$ mg/mL, 300 K, $\delta_{\text{pulse}} = 2$ ms, $\Delta = 285$ ms)

HES	$M_{\text{p,HES}}$, kDa	D , 10^{-10} m ² /s		HES	$M_{\text{p,HES}}$, kDa	D , 10^{-10} m ² /s	
		HES	water			HES	water
HETA 6K	5.5	1.169	20.2 ± 0.1	HETA 230K	201.0	0.305	20.1 ± 0.1
HETA 9K	8.9	0.986	20.2 ± 0.1	HETA 250K	219.3	0.302	20.2 ± 0.1
HETA 18K	16.1	0.731	20.2 ± 0.1	HETA 380K	330.2	0.241	20.2 ± 0.1
HETA 60K	52.8	0.548	20.1 ± 0.1	HETA 460K	396.9	0.239	20.2 ± 0.1
HETA 88K	80.0	0.415	20.1 ± 0.1	HETA 600K	468.7	0.197	20.1 ± 0.1
HETA 160K	153.6	0.356	20.2 ± 0.1	HETA 1300K	1004	0.142	20.2 ± 0.1

where c and α are numerical constants depending on the conditions of the experiment, the parameters of pulse gradient sequence, the size of the studied object, and the type of nonvalent interactions involving it.

The dependence of the D value on the size of molecular objects and the conditions of the experiment (temperature and viscosity of the solution) is described by the Stokes–Einstein equation [21]:

$$D = kT/6\pi\eta R_h,$$

where k is the Boltzmann constant, J/K; T is absolute temperature, K; η is the viscosity of the solution, N s/m²; and R_h is hydrodynamic radius, m.

The R_h value is composed of the radius of the diffusing particle itself and the thickness of the solvation shell. In the study of the self-diffusion of particles with the shapes distinct from spherical, R_h is usually interpreted as an average hydrodynamic radius of a solid sphere for which D in the given solution is equal to that of the studied object at all other conditions being equal.

The effect of selected pulse gradient sequence on the D value is described by the Stejskal–Tanner analytical function, which approximates the dependence of the amplitude of the spin echo signal on the parameters of the pulse of magnetic field gradient [34]:

$$I(\delta, \Delta) = I_0 \exp[-D\gamma^2 g^2 \delta_{\text{pulse}}^2 (\Delta - \delta_{\text{pulse}}/3)],$$

where I and I_0 are amplitudes of spin echo signal in the presence and absence of a pulse of magnetic field gradient; Δ is diffusion time, ms; δ_{pulse} is the duration of gradient pulse, ms; γ is the gyromagnetic ratio for the given nuclei, Hz/T; g is gradient of magnetic field, T/m.

For uncharged polysaccharides the following correlation equation was proposed [24]:

$$D = 8.2 \times 10^{-9} M_w^{-0.49} (M_w, \text{Da}; D, \text{m}^2/\text{s}). \quad (1)$$

The fact that Eq. (1) was derived on varying the parameters of pulse gradient sequence for each analyzed sample, $\Delta = 400$ – 1000 ms and $\delta_{\text{pulse}} = 1.0$ – 2.3 ms, i.e., the D values were fitted for a certain mathematical model, attracted our attention.

We used a quantitative dependence of MW on the HES D pulse sequence with the fixed values of Δ and δ_{pulse} for the whole series of analyzed HES samples (285 and 2 ms respectively). The parameters Δ and δ_{pulse} pulse used were selected in accordance with the following criteria:

(1) reproducibility of the water self-diffusion coefficient in polysaccharides solutions presented in publications (20.2×10^{-10} m²/s [24]);

(2) application of a pulse gradient sequence with the given parameters to HES in a wide range of molecular weights;

(3) resolution of HES with close MW values by D values in DOSY spectra.

The experimental D values of the components of various aqueous solutions of HES obtained under the constant conditions of the experiment and the parameters of pulse gradient sequence are presented in Table 1. The D values correspond to the peaks in their maxima (D_p) in the diagram of D distribution. For example, in Fig. 1 a diagram of the distribution of D of an aqueous HETA 18 solution with $D_p = 0.731 \times 10^{-10}$ m²/s is shown. The corresponding two-dimensional ¹H-DOSY spectrum is presented in Fig. 2, from which it is seen that signals of H(1) protons of the substituted and unsubstituted α -D-glucopyranose residues are observed in the region 5.20–5.50 ppm; the water signal ($D = 20.2 \times 10^{-10}$ m²/s) is present in the region of 4.65 ppm; signals of H(2)–H(6) protons of α -D-glucopyranose residues and four protons of hydroxyethyl substituents in the HES macromolecules appear in the region 3.2–3.9 ppm. Signals of different structural fragments of HETA18 have close D values (0.731×10^{-10} m²/s). As follows from the data in Table, the D values of water obtained in DOSY experiments with aqueous solutions of various HES remained virtually unchanged and equal to the values in the range of the error of experiment presented in publications, while D of HES decreased with an increase in molecular weight. A graphical dependence of the MW values presented in Table 1 on D is shown in Fig. 3. This dependence was studied using regression analysis. It

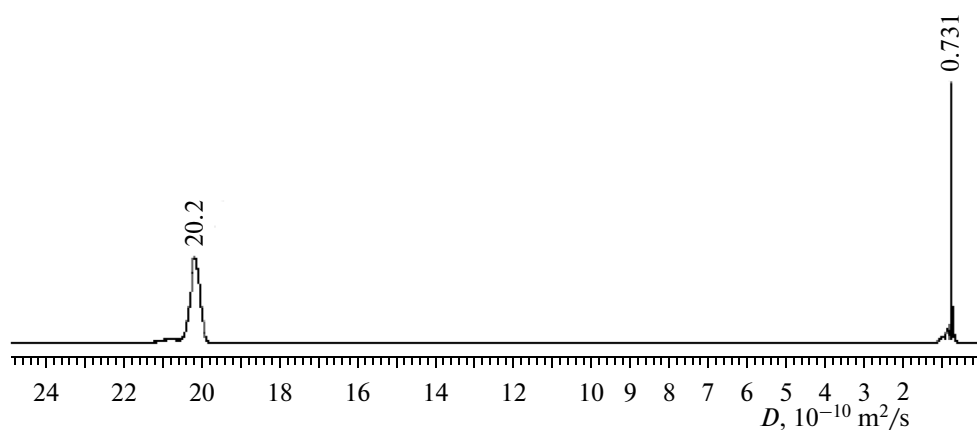


Fig. 1. Diagram of the distribution of self-diffusion coefficients of the components of an aqueous HES HETA18 solution.

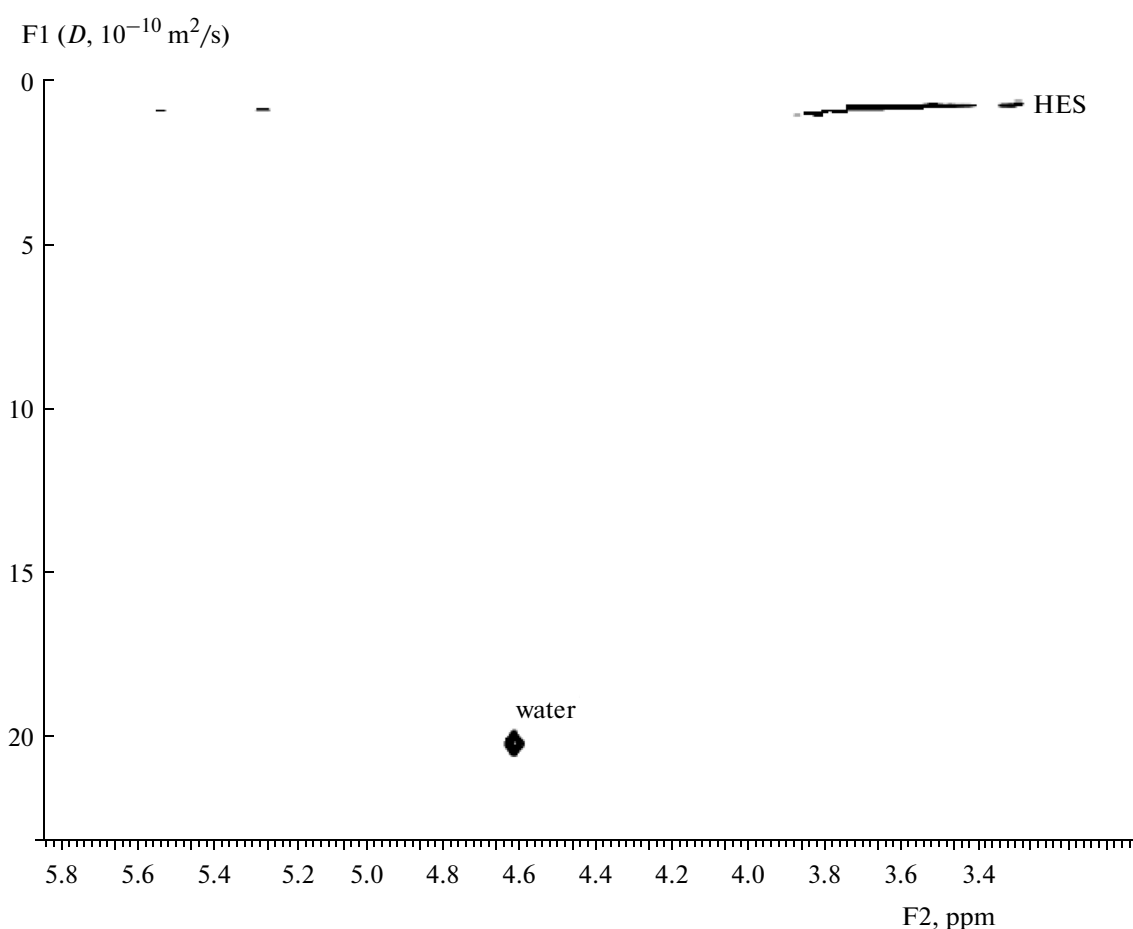


Fig. 2. A ^1H -DOSY spectrum of an aqueous HES HETA18 solution.

was found that the regression line (Fig. 3) was described by a power function:

$$\text{MW} = 9.0314D^{-2.5375} \quad (\text{MW, kDa}; D, 10^{-10} \text{ m}^2/\text{s}). \quad (2)$$

The correlation coefficient $R^2 = 0.990$. The power dependence can be reduced to the linear one using a

logarithmic transformation of the MW and D values. The linear regression model had the form:

$$\log \text{MW} = -2.5375 \log D - 24.419.$$

Let us compare Eq. (2) of the power pair regression of MW on D for HES with similar equations which we

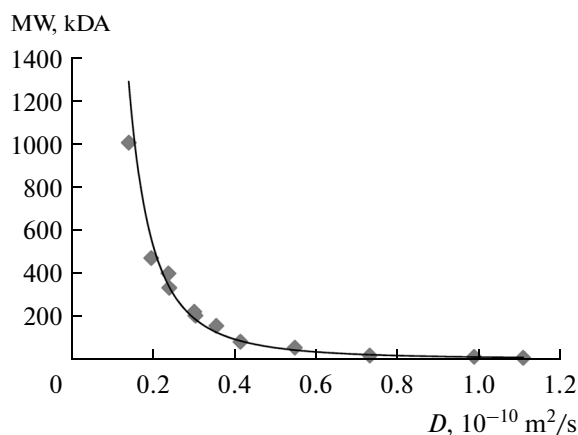


Fig. 3. Relationship between the experimental values of MWs and self-diffusion coefficients D of standard HES samples.

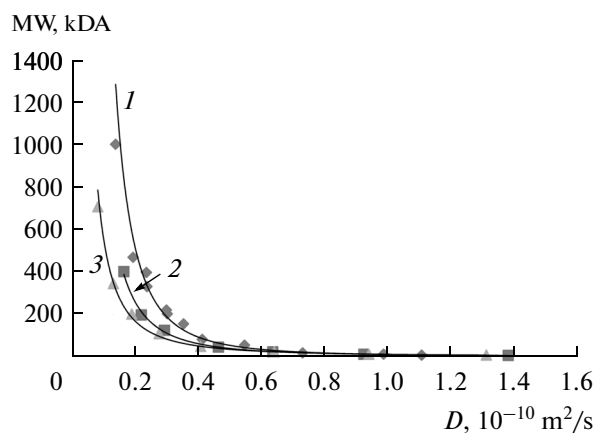


Fig. 4. Power dependencies $MW = cD^{-\alpha}$ for: (1) HES; (2) dextrans; and (3) pullulans.

obtained earlier for dextrans [35] and pullulans [36] (Table 2, Fig. 4). As it follows from Table 2, the absolute value of power index in the power functions increases in the series pullulans < dextrans < HES. Hence it follows that the constant D values of uncharged polysaccharides characterized by similar types of nonvalent interactions and different degrees of branching correspond to different MWs and that the MW values decrease in the series HES > dextrans > pullulans. This conclusion is in good agreement with the known data that branched polymers have smaller sizes compared to linear polymers [30]. It can be seen in Fig. 4 that significant differences in MW values are observed at $D < 0.6 \times 10^{-10} \text{ m}^2/\text{s}$. To illustrate this conclusion, MWs of polysaccharides with various degrees of branching calculated using the corresponding correlation equations (Table 2) are presented in Table 3. It follows from the data in Table 3 that the determination of an MW of HES based on the calibration of the experimentally measured diffusion parameter using the known molecular weights of dextrans or pullulans (common practice in using GPC and DOSY methods) leads to erroneous results for high-molecular HES (MWs greater than 50 kDa).

It should be noted that in the case of HES, in contrast to dextrans and pullulans, some scattering of the points of correlation field occurs relative to the theoretical regression line (Fig. 4). This is an evidence of the presence of an additional factor (in addition to the number of pyranose units in the macromolecule and

branching of the polymer chain) affecting the size of the polysaccharide macromolecule. We can suppose that the parameter MS is this factor depending on the conditions of HES synthesis and can vary over a wide range. Thus, all HES were classified according to the MS value as highly substituted ($MS = 0.62\text{--}0.75$), moderately substituted ($MS \sim 0.5$), and low substituted ($MS \leq 0.4$) [7]. The nonuniformity of the HES MS value can be responsible for an increase in the error of MW calculation using the equation of the pair power regression of MW on D . The calculated MW values for a series of HES standard samples not included into the learning sample are presented in Table 4. Calculation was performed using equations of power regression (1) and (2). The error of calculation (δ) was estimated relative to M_p values determined by GPC. As it follows from Table 4, the error of MW determination on the basis of equation of pair power regression (1) (learning sample is nonuniform by the degree of polysaccharide branching) is substantially higher than that using Eq. (2) (learning sample consists only of HES). Consequently, the consideration of polymer topology in deriving the correlation equation made substantially improved the accuracy of MW determination using the DOSY method. At the same time, the mean error of determining the MW of HES using Eq. (2) (15.49%) exceeds the mean error of determining the MW of dextrans (7.03%) [35]), which was comparable to the error of the DOSY method (not higher than 10% [37]). We can suppose that the trans-

Table 2. Equation of power regression of MW on D for polysaccharides with various degrees of branching

Regression equation (MW, kDa; D , $10^{-10} \text{ m}^2/\text{s}$)	Area of application	References
$MW = 9.0314D^{-2.538}$	Branched HES	The present work
$MW = 8.3038D^{-2.164}$	Low branched dextrans	[35]
$MW = 9.4862D^{-1.801}$	Linear pullulans	[36]

Table 3. Comparison of calculated MWs of polysaccharides with various degrees of branching and similar D values

D , 10^{-10} m ² /s	MW, kDa*			D , 10^{-10} m ² /s	MW, kDa		
	HES	dextran	pullulan		HES	dextran	pullulan
1	9.03	8.30	9.49	0.6	33.02	25.08	23.80
0.9	11.80	10.43	11.46	0.5	52.45	37.21	33.06
0.8	15.91	13.46	14.18	0.4	92.41	60.31	49.41
0.7	22.33	17.97	18.03	0.3	191.79	112.41	82.95

* Correlation equations from Table 2 were used for MW calculation.

Table 4. Comparison of MW_{HES} values calculated using Eqs. (1) and (2) with the experimental data obtained using GPC

HES	MW _{HES,exp} , kDa	D_{HES} , m ² /s	Equation (1)		Equation (2)	
			MW _{HES} , kDa	δ^* , %	MW _{HES} , kDa	δ , %
HETA 130K	121.1	0.378	58.61	51.6	106.68	11.91
HETA 175K	176.1	0.303	92.05	47.73	187.01	6.18
HETA 337K	374.0	0.263	122.89	67.14	267.86	28.38

* $\delta = ((\text{MW}(\text{calculated}) - \text{MW}(\text{experimental})) / \text{MW}(\text{experimental})) \times 100\%$.

fer from pair regression to multiple regression and the introduction of an MS parameter into the regression equation along with D will improve the accuracy of MW determination by the DOSY method. An advantage of this method is also the generation of spectral information (values of chemical shifts and multiplicities of signals) alongside with the diffusion data, which ensures the quantitative determination of the main characteristics of HES affecting their pharmacokinetic properties within one experiment.

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